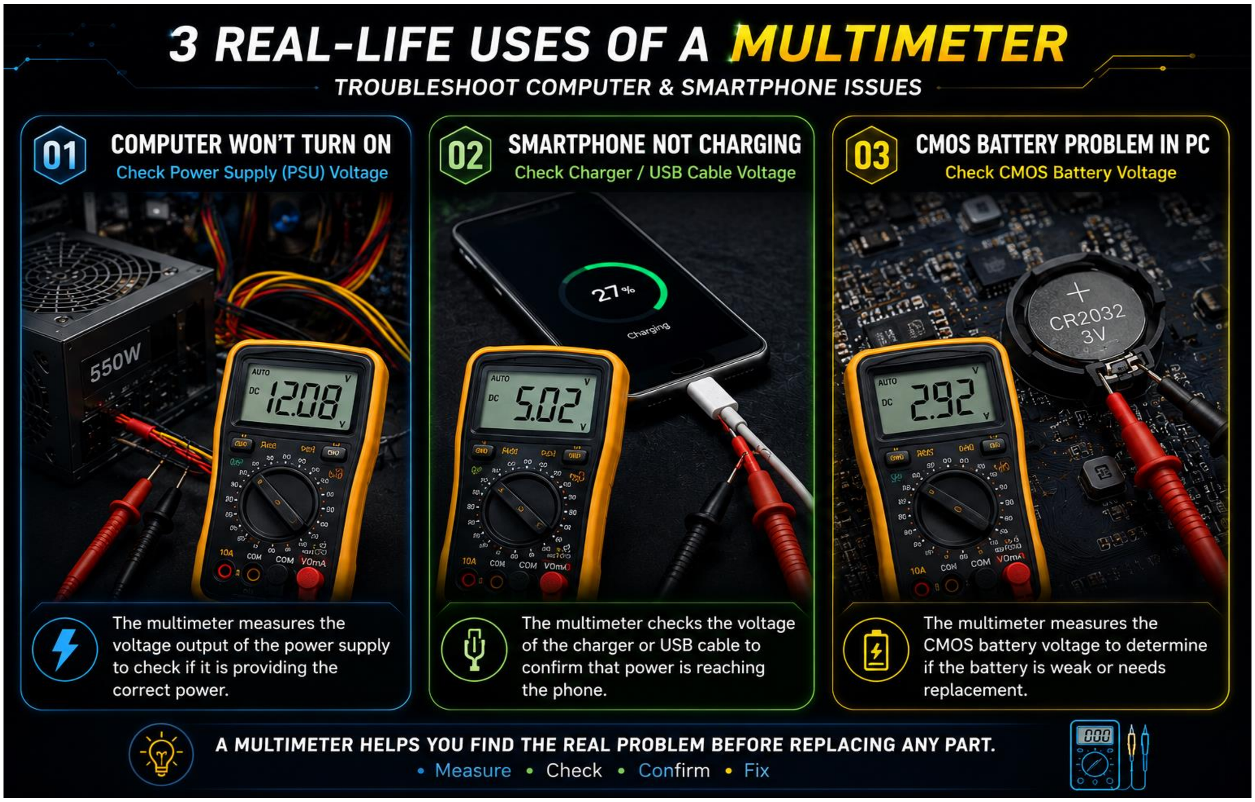
1 List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.



1. Computer won't turn on

    1. Measure the PSU (Power Supply Unit) voltage.
    2. Checks whether the computer is receiving correct power.

2. Smartphone not charging

    - ➢ Measure the charger or USB cable voltage.
    - ➢ Confirms that power is reaching the phone.

3. CMOS battery problem in PC

    - Measure the CMOS battery voltage.
    - Identifies if the battery is weak and needs replacement.

2 Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

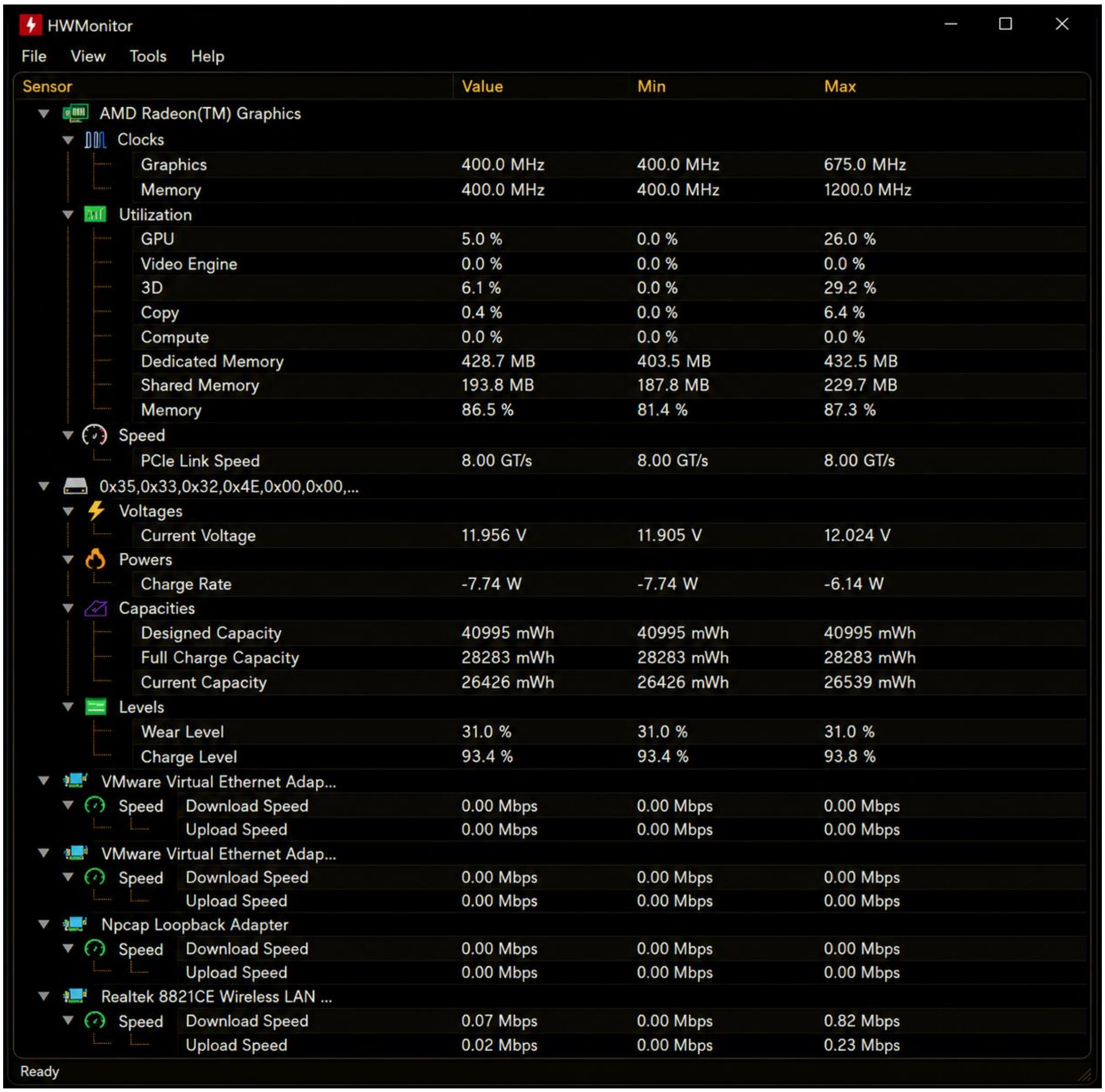
❖ POST CARD ::

- I watched a demo video of a POST Card used on a motherboard.
- The POST Card shows the hardware checking process done by the BIOS.
- It displays Hexadecimal Debug Codes such as C1, 55, and 80 on the LED screen.
- Each code shows a different hardware testing stage.
- In the demo, the final code was 80, which means the POST process finished successfully.

- These codes help identify RAM or motherboard boot problems.
- Therefore, a POST Card is a useful diagnostic tool for finding computer hardware faults.

3 Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

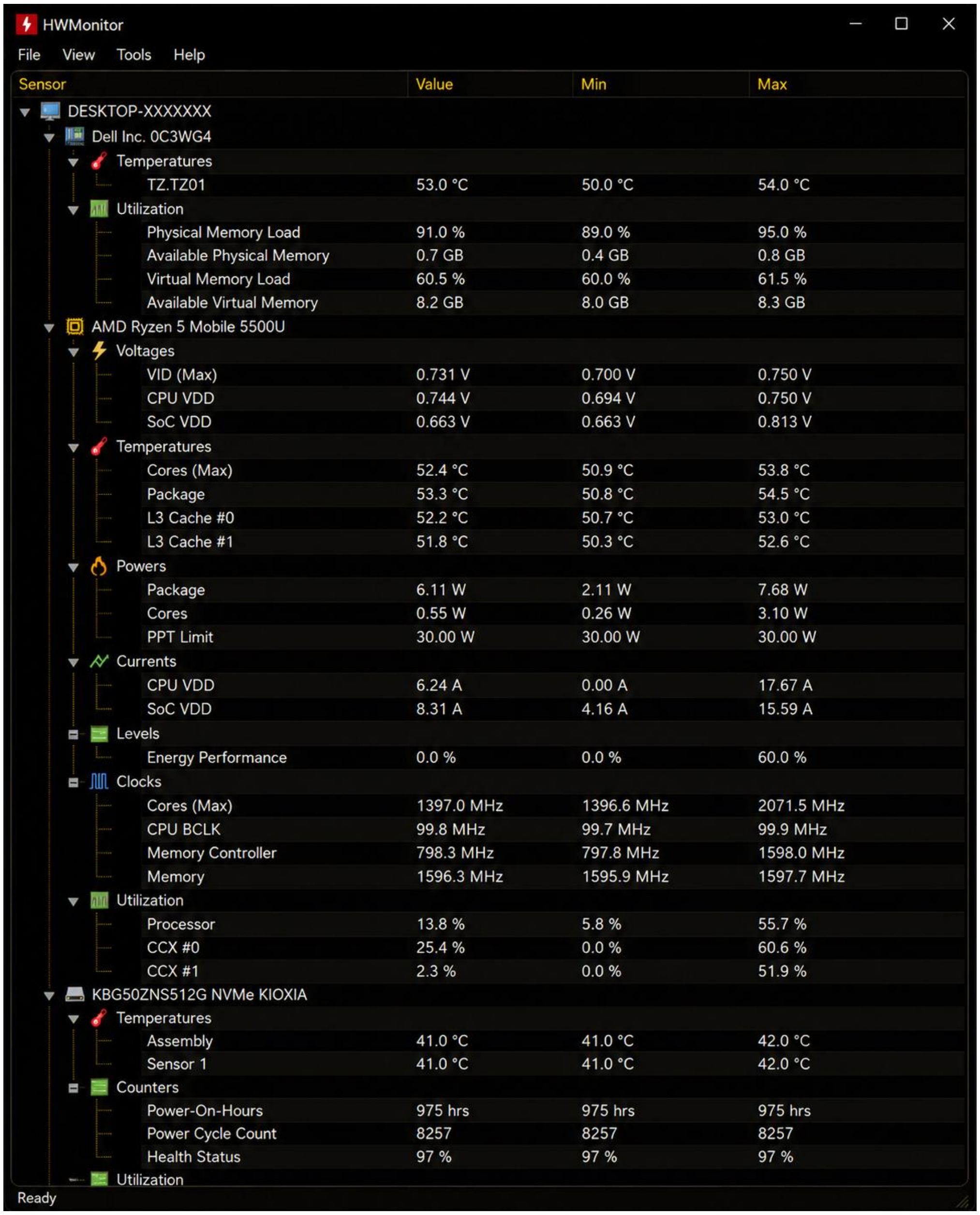


HWMonitor

File View Tools Help

Sensor	Value	Min	Max
AMD Radeon(TM) Graphics			
Clocks			
Graphics	400.0 MHz	400.0 MHz	675.0 MHz
Memory	400.0 MHz	400.0 MHz	1200.0 MHz
Utilization			
GPU	5.0 %	0.0 %	26.0 %
Video Engine	0.0 %	0.0 %	0.0 %
3D	6.1 %	0.0 %	29.2 %
Copy	0.4 %	0.0 %	6.4 %
Compute	0.0 %	0.0 %	0.0 %
Dedicated Memory	428.7 MB	403.5 MB	432.5 MB
Shared Memory	193.8 MB	187.8 MB	229.7 MB
Memory	86.5 %	81.4 %	87.3 %
Speed			
PCIe Link Speed	8.00 GT/s	8.00 GT/s	8.00 GT/s
0x35,0x33,0x32,0x4E,0x00,0x00,...			
Voltages			
Current Voltage	11.956 V	11.905 V	12.024 V
Powers			
Charge Rate	-7.74 W	-7.74 W	-6.14 W
Capacities			
Designed Capacity	40995 mWh	40995 mWh	40995 mWh
Full Charge Capacity	28283 mWh	28283 mWh	28283 mWh
Current Capacity	26426 mWh	26426 mWh	26539 mWh
Levels			
Wear Level	31.0 %	31.0 %	31.0 %
Charge Level	93.4 %	93.4 %	93.8 %
VMware Virtual Ethernet Adap...			
Speed Download Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
Upload Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
VMware Virtual Ethernet Adap...			
Speed Download Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
Upload Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
Npcap Loopback Adapter			
Speed Download Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
Upload Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps
Realtek 8821CE Wireless LAN ...			
Speed Download Speed	0.07 Mbps	0.00 Mbps	0.82 Mbps
Upload Speed	0.02 Mbps	0.00 Mbps	0.23 Mbps

Ready



- I downloaded and installed **HWMonitor**, a free software diagnostic tool, on my computer.

- I opened **HWMonitor** and checked the main dashboard.

- It showed important system health details such as **CPU temperature, CPU usage, voltage, power, GPU information, storage temperature, and SSD health status**.

- I checked the **Value, Minimum (Min), and Maximum (Max)** readings of different hardware components.

- I took screenshots of the HWMonitor dashboard showing these system details.

- Before submitting the screenshots, I **cropped or blurred any personal information** for privacy and safety.

4 Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it Checks	Example Problem Detected
Multimeter	Checks voltage, current, and resistance.	Low power supply voltage.
POST Card	Checks hardware during computer startup.	RAM or motherboard problem.
Software Diagnostics	Checks system and hardware health.	High CPU temperature or SSD health problem.